MILITARY SPECIFICATION

# HOLDER, FLIGHT CHECK LIST

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a scroll type flight check list holder.

1.2 Classification. Holders shall be of the following types and styles as specified (see 6.2).

## 2. APPLICABLE DOCUMENTS

*2.1 Government documents.

*2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging, Methods of |
| MIL-L-3661/1 | Lampholders, Styles LH20 and LH22 |
| MIL-G-3787 | Glass Laminated, Flat; (Except Aircraft) |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of US Military Property |
| MIL-STD-143 | Specifications and Standards Order of Precedence for the Selection of |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of |
| MS25036 | Terminal, Lug, Crimp Style, Copper, Insulated, Ring-Torque, Bell-Mouthed, Type Ii, Class I |
| MS25231 | Lamp, Incandescent, Single Contact, Miniature Bayonet Base, T-3-1/4 Bulb (Asg) |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this documet or by letter.

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| | |
|---|---|
| MS91528 | Knobs-Control, Plastic (Round, Concentric, Pointer, Spinner, Spinner Slip Clutch, Bar, Tactile, Knob Lock Pointer, and Knob Locks) |
| MS91531 | Knob, Control, Round, Plastic (With Fluted Grip) |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

*2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the test of this specification shall take precedence.

3. REQUIREMENTS

3.1 Components. The flight check list holder shall consist of a case, lens and lens frame, two rollers with knobs, light, light dimming rheostat and switch, fuse plate, mounting bracket, and electrical connections.

3.2 Dimensions. The dimensions which are controlled by this document are listed in applicable paragraphs of Section 3 and Figure 1. Other dimensions are controlled by the manufacturer's design.

3.3 Materials. Materials used in the construction of the holder shall be of uniform quality and condition and free of seams, cracks, or other defects which may adversely affect the appearance, strength, endurance, or wear resistance. Materials shall not be filled or otherwise treated to conceal defects.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.4.1 and 3.4.2.

3.4.1 Standard parts. AN or MS standard parts shall be used where they are suitable for the prupose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are interchangeable with AN, MS or MIL Standard parts without alterations.

3.4.2 Commercial parts. In applications for which no suitable corresponding AN, MS, or MIL type part is in effect on the date of inviatation for bids, commercial parts may be used provided they conform to the requirements of this document.

3.5 Design and performance. The flight check list holder shall be designed and constructed so that no parts will work loose in service. All parts subject to wear, breakage, or distortion shall be accessible for adjustment or replacement. All parts shall be capable of withstanding continued usage for the purposes intended. The holder shall move the scroll type check list across a glass lens when a knob on one roller is turned. A light shall illuminate the entire width of the check list without bright spots.



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3.5.1 Holder capacity. The holder shall have a minimum capacity of eight (8) feet of scroll check list 2 13/16 inches wide (when trimmed) and 0.005 inch thick.

3.6 Details of components. The flight check list holder shall be furnished as a completely assembled unit. Any components required for operation of the unit shall be included even though they may not be listed herein.

3.6.1 Case. The case shall be constructed from thermal setting or phenolic plastic with high impact properties. Other plastics with equal or better properties will be acceptable. The case shall serve as a frame for the mounting bracket, light, rollers, rheostat with switch, electrical input terminal and fuse plate. The case dimensions shall conform to those of Figure 1. The case shall have provisions for installing the mounting bracket on either end so that the holder may be used as a left hand or a right hand model.

*3.6.1.1 Case and lens. The case shall have tabs molded into the ledge of the body for support of the lens. This will preclude the lens from contacting the fuse plate.

*3.6.2 Lens frame and lens. The lens frame shall have a glass lens (see 3.6.2.2) conforming to the dimensions in Figure 1 and placed symmetrically about the center lines of the frame. The lens frame and lens shall be so constructed that when they are mated, lateral movement between the lens frame and lens shall be a minimum. The lens shall not contact the fuse plate during lateral movement. The lens frame shall be retained to the case with four (4) screws.

3.6.2.1 Lens frame. The lens frame shall conform to the material and color of the case.

*3.6.2.2 Lens. The lens shall be made of flat laminated glass, class I, type A, in accordance with MIL-G-3787. The edges of the glass shall be ground, or otherwise treated to remove roughness and nicks. The lens shall have a yellow indicating line 0.010 to 0.015 inch wide across the center (see Figure 1). This line is to be etched into the lens approximately 0.005 $\pm$ 0.002 deep and when assembled should be on the inside of the holder.

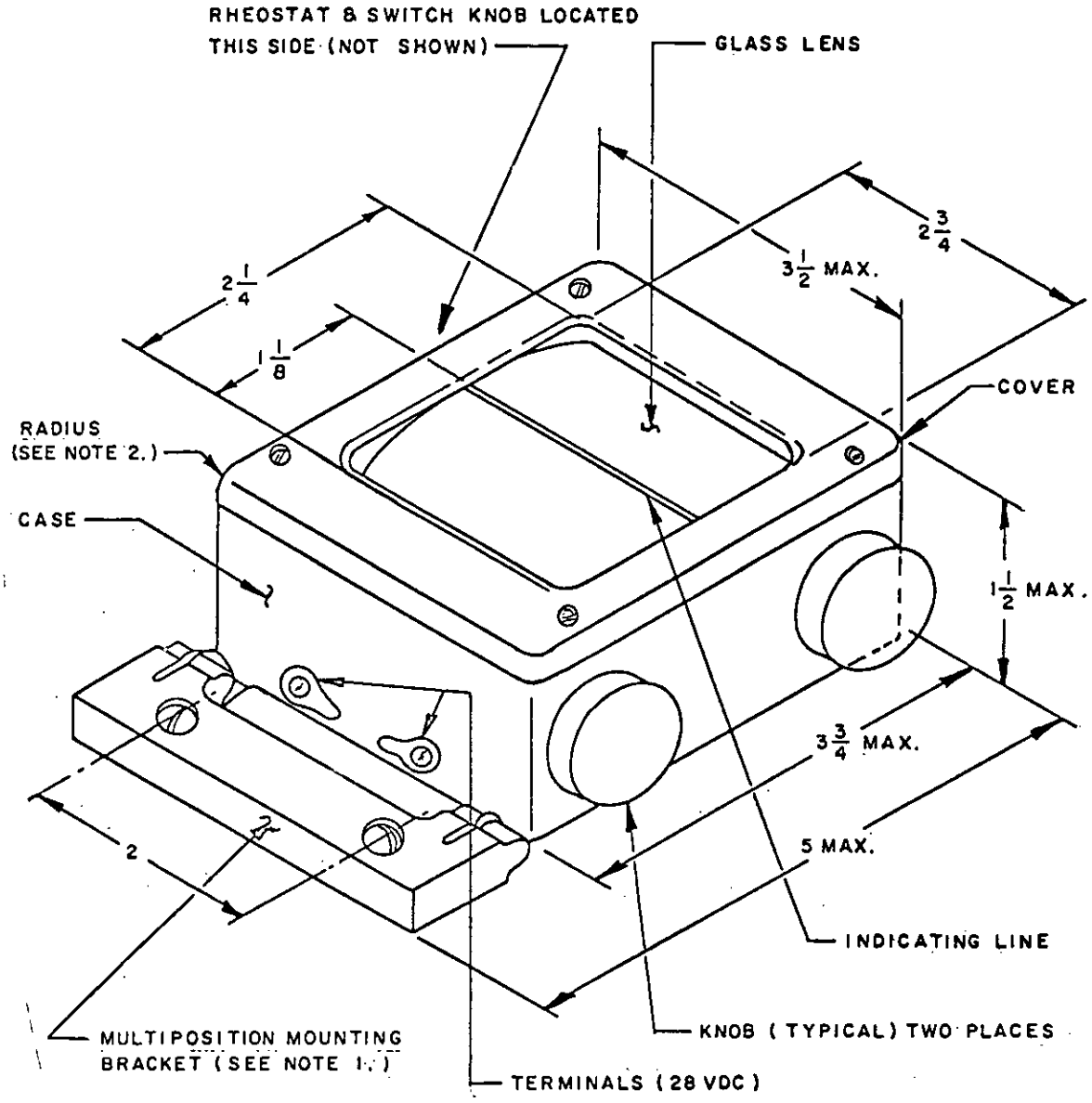
*3.6.3 Rollers. Each roller shall be sized to hold the entire scroll check list required by 3.5.1. The rollers shall have a shaft which extends through the case on one side for attaching a turning knob. Both rollers shall be provided with a mechanical means of attaching one end of the scroll check list. The check list shall not become detached when the check list is rolled to the end on the other roller. Tape or tacks for attaching the check list to the roller will not be acceptable.

3.6.3.1 Knobs. A knob shall be provided to rotate each roller. The Knobs shall be type MS91531-2N2B.

3.6.3.2 Lamp knob. The lamp rheostat shall be provided with a knob type MS91528-1N2B.

3.6.3.3 Spring. A wavy spring shall be installed between the case and roller or the knob and case to keep the scroll taut.

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NOTES:

1. MOUNTING BRACKET TO ROTATE FROM POSITION SHOWN UPWARD TO A POSITION PARALLEL WITH THE END OF THE CASE. DOWNWARD ROTATION OPTIONAL.
2. CORNERS AND EDGES OF CASE AND COVER TO HAVE A MINIMUM 1/16 INCH RADIUS.
3. DIMENSIONS ARE IN INCHES $\pm$ 1/64.
4. MAXIMUM WEIGHT 1.1 LBS.

FIGURE 1

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3.6.4 Fuse plate. The fuse plate shall be made of colorless frosted glass or plastic. The fuse plate shall not distort from the heat of the light. The fuse plate shall be approximately the same size as the lens opening. The fuse plate will be of a constant cross section, except for the beveled edges over which the check list rolls. It shall be notched to fit into the tabs of the case, and shall be lower than the top of the tabs so as to comply with 3.6.4.1.

3.6.4.1 Fuse plate position. The fuse plate shall be positioned in the case so that the scroll is not in continous contact with the lens. The distance between the lens and fuse plate shall utilize the maximum space permissible by the manufacturer's design.

3.6.5 Light. A red lamp per MS25231-313R or green phosphoresce lamp per ILSAA (Improved Lighting System for Army Aircraft) and lampholder, type LH20, in accordance with MIL-L-3661/1 shall be installed in the case under the fuse plate. The bracket mounting lampholder shall be positioned so that the lamp filament is in the center of the case.

3.6.6 Rheostat and switch. A rheostat shall be provided to dim the light,. The rheostat shall have a built-in switch for cutting the light off.

3.6.6.1 Rheostat mounting. The rheostat shall be mounted inside the case with a type MS91528-1N2B knob to control it on the outside of the case. The knob shall be on the side of the case opposite the roller rotating knobs.

3.6.7 Wiring. The unit shall be completely wired for 28 volts, direct current. Terminals for the electrical input shall be on the end of the case. Terminals shall accept MS25036 type terminal lugs.

*3.6.8 Mounting bracket. The mounting bracket shall be a multi-position type which will have a minimum of three positive positions. The bracket shall have a positive means for maintaining each position. The holder must maintain all selected positions upon experiencing 10 pounds of force applied at the center of the basic mold and normal (perpendicular) to the face or back of the holder. The bracket shall have two 3/16 inch diameter holes for mounting screws on 2 plus or minus 1/32 inch centers placed symmetrically about the centerline of the coordinator. The holes shall be counterbored to 9/32 plus 1/32 minus 0.00 inch to depth of 5/32 inch. The counterbored holes shall be on the face of the mounting bracket that folds against the end of the case.

3.6.8.1 The mounting bracket shall be made of aluminum or aluminum alloy. Aluminum or aluminum alloy shall be anodized in accordance with MIL-A-8625.

3.6.9 Color. The color of the case shall be flat black.

3.7 Identification of product. Assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship. Standards of workmanship shall be such that holder assembly specified herein shall comply with requirements in Section 3 and test in Section 4. Workmanship shall be of such quality as to assure a product free of defects such as materials, parts, freedom from binding, freedom of parts from burrs an sharp edges, defective connections, loose, damaged or missing parts and finishes.

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*3.9 Reclaimed material. The use of reclaimed material shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the flight check list holder shall consist of the acceptance tests identified in 4.3.

4.3 Acceptance test. Acceptance test shall consist of the following:

a. Individual test (see 4.3.1)
b. Sampling test (see 4.4)

4.3.1 Individual test. Each holder submitted for acceptance shall be examined using the following list of defects (see 4.3.1.1). If the holder has one or more defects it shall be rejected.

*4.3.1.1 List of defects:

a. Component missing.

b. Light does not illuminate.

c. Light does not dim properly.

d. Cracked or broken lens.

e. Cracked or broken cover.

f. Cracked or broken fuse plate.

g. Cracked or broken case.

h. Materials not as specified.

i. Mounting bracket does not maintain each positive position. (see 3.6.8).

j. Workmanship not acceptable.

k. Dimensions not as specified.

l. Rollers fail to turn smoothly (more drag than necessary to maintain scroll position).

m. Exceeds maximum weight.

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4.3.2 Cycle test. Rollers shall be capable of withstanding 2,000 cycles of operation (full movement of the 8 foot chart) with no measurable increase in side-play or end-play of the rollers.

4.4 Sampling test. Samples shall be selected at random from each inspection lot and subjected to individual test and tested for compliance with 3.6.2.2 and 3.6.3.1. The number of samples shall be selected according to 4.4.1. Failure to meet the requirements shall be cause for rejection of the entire lot.

4.4.1 Number of samples.

| LOT SIZE | NUMBER OF SAMPLES |
|---|---|
| 25 or less | 1 |
| 26 to 50 | 2 |
| 51 to 100 | 3 |
| 100 or more | 3 plus 1 for each 100 or fraction thereof |

4.5 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction all sampling test shall be repeated.

4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

*5 PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, each holder shall be individually preserved and packaged in accordance with MIL-P-116, Method 1C1 without preservative compound. Insofar as practicable each package shall be uniform in shape and size and shall be designed to enclose the contents in a compact manner.

5.1.2 Level C. Preservation and packaging that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.



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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, each holder shall be packed in accordance with MIL-STD-794. Shipping container shall be exterior export type conforming to Tales II, III or IV of MIL-STD-794.

5.2.2 Level B. Unless otherwise specified, each holder shall be packed in accordance with MIL-STD-794. Shipping container shall be exterior domestic type conforming to Tables II, III or IV of MIL-STD-794.

5.2.3 Level C. Shipping containers shall meet the requirements of MIL-STD-794 for level C packing.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This flight check list holders covered by this document are to be used for holding a scroll check list on transport and helicopter type aircraft.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Sampling and inspection procedures required (see 4.4).

c. Level of preservation and packaging (see 5.1).

d. Level of packing (see 5.2).

*6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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